Design and Analysis of Class EF2 Inverter

Ankit Singh
Department of Electrical Engineering
IIT BHU, Varanasi, India

***Abstract*—The increasing demand for energy-efficient power electronics has driven research into high-frequency inverter topologies for applications such as wireless power transfer (WPT), RF systems, and resonant DC–DC converters. The Class EF2 inverter emerges as an advanced topology that combines the advantages of Class E and Class F inverters, offering zero-voltage switching (ZVS), reduced switching losses, and lower voltage stress. This paper presents an overview, circuit operation, and waveform characteristics of the Class EF2 inverter.**

I. Introduction

The increasing demand for energy-efficient power electronics has driven research into high-frequency inverter topologies for applications such as wireless power transfer (WPT), radio-frequency (RF) systems, and resonant DC–DC converters. These applications require inverters that can operate at high frequencies with low switching losses, reduced component stress, and high efficiency.

Conventional inverter topologies such as Class D, Class E, and Class F are widely used. Class D suffers from high switching losses due to hard switching. Class E enables zero-voltage switching (ZVS) but experiences high peak voltage stress. Class F reduces switching losses using harmonic tuning but increases design complexity.

To address these challenges, the Class EF2 inverter incorporates a second-harmonic resonant network into the Class E structure, achieving improved efficiency and reduced device stress.

II. Class EF2 Inverter: Overview

The Class EF2 inverter is an advanced switched-mode power amplifier combining features of Class E and Class F topologies. It retains zero-voltage switching (ZVS) while incorporating second-harmonic tuning.

A parallel resonant network tuned to the second harmonic is added to the conventional Class E circuit. This network shapes the drain voltage waveform and reduces peak switch voltage.

Unlike Class E, where peak voltage reaches approximately $3.56V_{DD}$, the Class EF2 inverter limits it to about 2–$3V_{DD}$, reducing device stress and improving efficiency.

III. Circuit Diagram

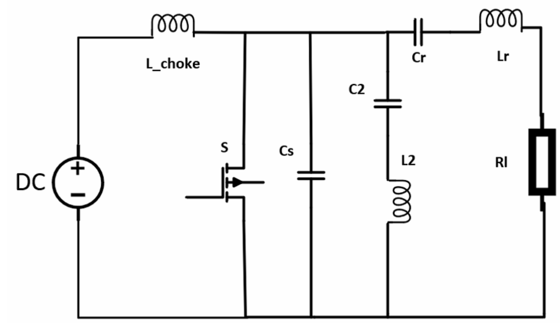

Fig. 1. Circuit diagram of the Class EF2 inverter

IV. Circuit Components and Operation

A. DC Supply and Feed Inductor (L_{choke})

A large RF choke connects the DC supply V_{DD} to the drain. It provides high impedance at switching frequency, ensuring nearly constant current.

B. Switching Device

A MOSFET or GaN HEMT operates as a switch driven at frequency f_0 with duty cycle $D \approx 0.4$–0.5.

C. Shunt Capacitance (C_s)

This includes intrinsic capacitance C_{oss} and shapes the voltage waveform, enabling ZVS.

D. Second Harmonic Tank ($L_2 - C_2$)

This parallel resonant circuit is tuned to $2f_0$ and blocks second harmonic current, modifying the voltage waveform.

E. Load Network ($L_r - C_r - R_L$)

A series resonant network tuned to f_0 ensures sinusoidal current delivery to the load.

V. Zero Voltage Switching (ZVS)

For ideal ZVS operation:

$$v_{DS}(\omega t = 2\pi) = 0 \tag{1}$$

$$\frac{dv_{DS}}{d(\omega t)} = 0 \quad \text{at } \omega t = 2\pi \tag{2}$$

These conditions ensure zero switching loss at turn-on.

VI. Operation of Class EF2 Inverter

A. On-State ($0 \leq \omega t < 2\pi D$)

The transistor conducts and $v_{DS} \approx 0$. Current flows through the switch.

B. Off-State ($2\pi D \leq \omega t < 2\pi$)

The switch is off, and the choke current charges the capacitance. The drain voltage rises and returns smoothly to zero.

VII. Role of Second Harmonic Tank

The second-harmonic tank is tuned to:

$$f_2 = 2f_0 \tag{3}$$

It confines second harmonic current and shapes the drain voltage waveform, reducing peak voltage.

This reduces voltage stress from $3.56V_{DD}$ (Class E) to approximately 2–$3V_{DD}$.

VIII. Mathematical Analysis and Design Equations

A rigorous mathematical framework forms the foundation upon which the Class EF2 inverter is designed and optimized. This section develops the analytical model of the circuit under steady-state operating conditions. Beginning from first principles and applying circuit laws, the voltage and current waveforms are derived, followed by the formulation of component design equations.

The analysis proceeds under the following standard idealizing assumptions:

1) The switching device operates as an ideal switch with zero on-resistance and infinite off-resistance.
2) The DC feed inductance L_{choke} is sufficiently large so that the supply current I_{in} remains constant over a switching cycle.
3) The series load network (L_r–C_r–R_L) has a high quality factor, ensuring sinusoidal output current at the fundamental frequency.
4) The shunt capacitance C_s is linear and constant, representing the transistor output capacitance.
5) All passive components are assumed lossless.

A. Time-Domain Analysis of Class EF2 Inverter

Let the normalized time variable be defined as:

$$n = \omega t \tag{4}$$

where $\omega = 2\pi f_0$ is the angular switching frequency.

The normalized inductor current is defined as:

$$y(n) = \frac{i_L(n)}{I_{in}} \tag{5}$$

The output current is assumed sinusoidal:

$$i_o(n) = I_m \sin(n + \phi) \tag{6}$$

The switching operation is divided into two intervals:

1) ON Interval ($0 \leq n < 2\pi D$): During this interval, the switch is closed and the voltage across it is zero. Applying Kirchhoff's Current Law (KCL) at the switching node:

$$i_s(n) = I_{in} - i_L(n) - i_o(n) \tag{7}$$

Since the second-harmonic tank is source-free during this interval, the inductor current can be expressed as:

$$\frac{i_L(n)}{I_{in}} = A_1 \cos(2n) + B_1 \sin(2n) \tag{8}$$

2) OFF Interval ($2\pi D \leq n < 2\pi$): During the off interval, the switch is open ($i_s = 0$). Applying KCL:

$$i_L(n) = I_{in} - i_o(n) - i_C(n) \tag{9}$$

The capacitor current is given by:

$$i_C(n) = \omega C_s \frac{dv_s(n)}{dn} \tag{10}$$

The switch voltage is obtained as:

$$v_s(n) = \omega L_2 \frac{di_L(n)}{dn} + \frac{1}{\omega C_2} \int i_L(n)\, dn \tag{11}$$

Differentiating:

$$\frac{dv_s(n)}{dn} = \omega^2 L_2 C_1 \frac{d^2 i_L(n)}{dn^2} + \frac{1}{\omega^2 C_2} i_L(n) \tag{12}$$

Substituting and normalizing:

$$\omega^2 L_2 C_1 \frac{d^2 y(n)}{dn^2} + \left(1 + \frac{C_1}{C_2}\right) y(n) = 1 - \frac{I_m}{I_{in}} \sin(n+\phi) \tag{13}$$

Dividing:

$$\frac{d^2 y(n)}{dn^2} + q^2 y(n) = \frac{1 - \frac{I_m}{I_{in}} \sin(n + \phi)}{\omega^2 L_2 C_1} \tag{14}$$

where

$$q^2 = \frac{C_1 + C_2}{\omega^2 L_2 C_1 C_2} \tag{15}$$

The general solution is:

$$y(n) = A_2 \cos(qn) + B_2 \sin(qn) - \frac{p^2}{q^2 - 1} \sin(n+\phi) + \frac{1}{k+1} \tag{16}$$

where

$$k = \frac{C_1}{C_2}, \quad p = \frac{1}{k+1} \frac{I_m}{I_{in}} \tag{17}$$

3) Boundary Conditions: Continuity conditions:

$$i_L(2\pi D^-) = i_L(2\pi D^+) \tag{18}$$

$$\left.\frac{di_L}{dn}\right|_{2\pi D^-} = \left.\frac{di_L}{dn}\right|_{2\pi D^+} \tag{19}$$

Soft-switching conditions:

$$v_s(2\pi D) = 0 \quad \text{(ZVS)} \tag{20}$$

$$\left.\frac{dv_s}{dn}\right|_{2\pi D} = 0 \quad \text{(ZVDS)} \tag{21}$$

B. Numerical Solution of Boundary Conditions

The six boundary conditions derived in the previous subsection form a system of nonlinear equations in the unknown coefficients. Due to the complexity of these equations, a closed-form analytical solution is not feasible. Therefore, a numerical approach is adopted.

An iterative numerical technique based on the Levenberg–Marquardt algorithm is employed to solve the nonlinear system. This method combines the advantages of gradient descent and the Gauss–Newton method, making it well-suited for solving nonlinear least-squares problems.

The iterative process is initialized with physically reasonable estimates for the unknown variables. Convergence is verified by ensuring that the residuals of all boundary conditions are minimized within a specified tolerance.

The operating point corresponding to optimal inverter performance is obtained for the following parameter values:

$$k = 0.867, \quad D = 0.375 \tag{22}$$

The converged numerical solution yields the following coefficients:

$$A_1 = 1.2405, \quad B_1 = 0.9394 \tag{23}$$

$$A_2 = -1.1091, \quad B_2 = 1.0073 \tag{24}$$

$$p = -1.2904, \quad \phi = 1.3569 \text{ rad} \tag{25}$$

These coefficients completely define the normalized inductor current, capacitor current, and switch voltage waveforms derived in the previous subsection. Using these values, all design parameters of the Class EF2 inverter can be determined analytically.

C. Power Relations and Design Coefficients

With the numerically obtained value of p and the known parameter k, the ratio of the sinusoidal output current amplitude to the DC supply current is given by:

$$\frac{I_m}{I_{in}} = p(k+1) = -1.2904 \times (0.867 + 1) = -3.5853 \tag{26}$$

The negative sign indicates a phase relationship and does not affect the magnitude of power calculations.

1) DC Input Resistance: The power balance between the DC source and load is expressed as:

$$P_{in} = V_{in} I_{in} = \frac{1}{2} I_m^2 R_L \tag{27}$$

Rearranging:

$$\frac{R_{DC}}{R_L} = \frac{1}{2}\left(\frac{I_m}{I_{in}}\right)^2 = \frac{1}{2}(3.5853)^2 = 6.427 \tag{28}$$

2) Normalized Output Power: The normalized output power is:

$$\frac{P_o R_L}{V_{in}^2} = \frac{1}{2}\left(\frac{I_m}{I_{in}}\right)^2 \left(\frac{R_L}{R_{DC}}\right)^2 = 0.1556 \tag{29}$$

3) Power-Output Capability Coefficient: The power-output capability coefficient is defined as:

$$c_p = \frac{P_o}{V_{s,peak} \cdot I_{s,peak}} \tag{30}$$

At optimum operation:

$$\frac{V_{s,peak}}{V_{in}} = 2.3162, \quad \frac{I_{s,peak}}{I_{in}} = 3.2632 \tag{31}$$

Thus,

$$c_p = 0.1323 \tag{32}$$

This represents a significant improvement over the classical Class E inverter, confirming enhanced power capability of the Class EF2 topology.

D. Design Parameter Relationships

With the DC input resistance established, all component values can now be derived analytically.

1) Step 1: Shunt Capacitance C_1: The shunt capacitance is given by:

$$C_1 = \frac{1}{7.5851\,\omega R_L} \tag{33}$$

2) Step 2: Second-Harmonic Capacitance C_2: Using the capacitance ratio:

$$k = \frac{C_1}{C_2} = 0.867 \tag{34}$$

$$C_2 = \frac{C_1}{0.867} \tag{35}$$

3) Step 3: Second-Harmonic Inductance L_2: The second-harmonic tank must satisfy:

$$f_2 = 2f_0 \quad \Rightarrow \quad \omega_2 = 2\omega \tag{36}$$

$$L_2 = \frac{1}{(2\omega)^2 C_2} \tag{37}$$

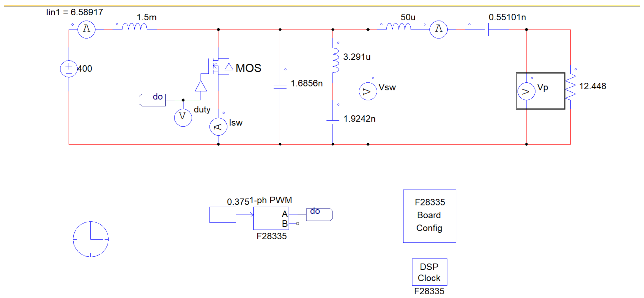

Fig. 2. PSIM-based simulation model of the designed Class EF2 inverter using calculated component values

4) Step 4: Series Load Network ($L_r - C_r$): The resonant condition is:

$$\omega L_r - \frac{1}{\omega C_r} = 2.0339 R_L \tag{38}$$

Typically, C_r is selected first based on quality factor, and L_r is calculated accordingly.

5) Step 5: DC Feed Inductance L_1: The minimum choke inductance required is:

$$L_{1,\min} = \frac{24.1024 R_L}{\omega} \tag{39}$$

In practice, L_1 is chosen to be 5–10 times this value to reduce ripple.

IX. Calculated Parameter Values for Class EF2 Inverter

Using the normalized design parameters obtained in the previous section, the actual component values for the Class EF2 inverter are calculated. The design specifications are:

- Output Power, $P_o = 500$ W
- Input Voltage, $V_{in} = 200$ V
- Switching Frequency, $f = 100$ kHz

The angular frequency is:

$$\omega = 2\pi f = 6.2832 \times 10^5 \text{ rad/s} \tag{40}$$

The load resistance is calculated using normalized power relation:

$$R_L = \frac{0.1556\, V_{in}^2}{P_o} = 12.448\ \Omega \tag{41}$$

The input current is:

$$I_{in} = \frac{P_o}{V_{in}} = 2.5 \text{ A} \tag{42}$$

The equivalent DC input resistance:

$$R_{DC} = 6.427\, R_L = 80\ \Omega \tag{43}$$

TABLE I
Complete Component Design for Class EF2 Inverter

Parameter	Expression	Value
P_o	Given	500 W
V_{in}	Given	200 V
f	Given	100 kHz
ω	$2\pi f$	6.2832×10^5 rad/s
R_L	$\frac{0.1556 V_{in}^2}{P_o}$	12.448 Ω
I_{in}	$\frac{P_o}{V_{in}}$	2.50 A
R_{DC}	$6.427 R_L$	80.00 Ω
C_1	$\frac{1}{7.5851 \omega R_L}$	16.86 nF
C_2	$\frac{C_1}{0.867}$	19.44 nF
L_2	$\frac{1}{(2\omega)^2 C_2}$	32.57 μH
Q_L	Chosen	10
L_r	$\frac{Q_L R_L}{\omega}$	198.1 μH
C_r	$\frac{1}{\omega Q_L R_L}$	12.79 nF
$L_{1,\min}$	$\frac{24.1024 R_L}{\omega}$	477.5 μH

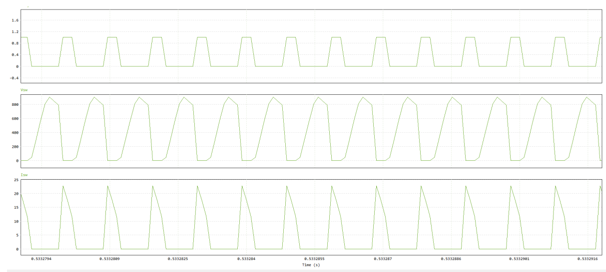

Fig. 3. ,Duty cycle,drain to source voltage and Drain current, waveform

X. Simulation Results and Discussion

The designed Class EF2 inverter is simulated using PSIM to validate the theoretical analysis and calculated design parameters. The simulation results demonstrate the switching behavior, voltage stress, and output characteristics of the inverter.

A. Switch Voltage Waveform (V_{DS})

The drain-to-source voltage waveform shows that the voltage across the switch returns to zero before turn-on, satisfying the Zero-Voltage Switching (ZVS) condition. This significantly reduces switching losses and improves efficiency.

B. Drain Current Waveform

The drain current waveform is non-overlapping with the voltage waveform, indicating soft-switching operation. The current flows primarily during the on-state of the switch, minimizing power dissipation.

C. Output Voltage Waveform

The output voltage waveform is nearly sinusoidal, confirming that the series resonant load network effectively filters higher-order harmonics.

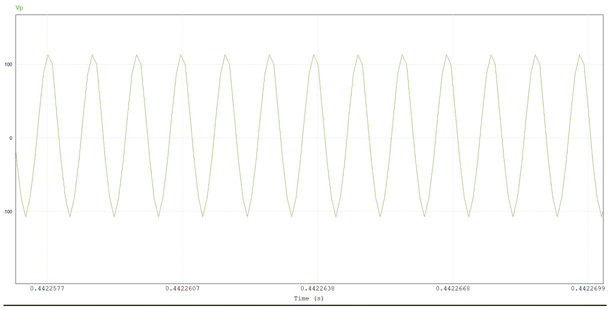

Fig. 4. Output voltage waveform across the load

D. Discussion

The simulation results validate the theoretical design of the Class EF2 inverter. The presence of ZVS operation ensures reduced switching losses, while the second-harmonic resonant tank limits peak voltage stress across the switch.

Compared to conventional Class E inverters, the Class EF2 topology achieves:

- Lower peak switch voltage stress
- Improved power-output capability
- Better efficiency at high switching frequencies

These results confirm that the Class EF2 inverter is suitable for high-frequency and high-efficiency power conversion applications.

XI. Post-Inverter Power Processing and DC Output

To obtain a usable DC output from the high-frequency AC output of the Class EF2 inverter, an additional stage consisting of a high-frequency transformer and rectifier circuit is implemented.

A. Isolation Using High-Frequency Transformer

A transformer with a turns ratio of 1:2 is connected at the output of the inverter. The purpose of this transformer is twofold:

- To provide electrical isolation between the inverter and the load
- To step down the voltage level suitable for the rectification stage

The high-frequency operation of the inverter allows the use of a compact transformer, reducing size and improving power density.

B. Rectifier Circuit and DC Output

The transformer output is fed into a diode-based rectifier circuit, which converts the high-frequency AC into DC. A filter capacitor is used at the output to smooth the rectified voltage.

A resistive load of $5\,\Omega$ is connected across the DC output to analyze the performance of the system.

C. Complete Circuit Configuration

The above figure shows the complete configuration, where the inverter output is processed through the transformer and rectifier to obtain a DC output voltage (V_{dc}) across the load.

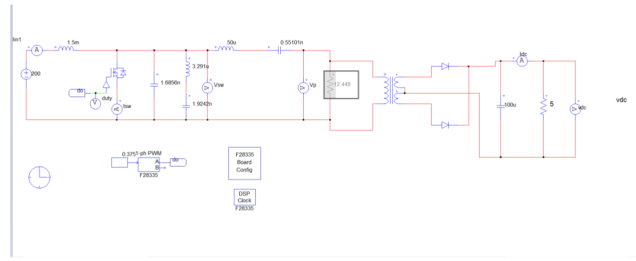

Fig. 5. Complete system including Class EF2 inverter, transformer (2:1), rectifier, and DC load

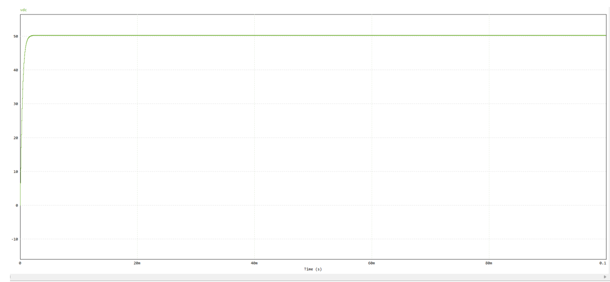

Fig. 6. DC output voltage (V_{dc}) across load

D. Discussion

The addition of the transformer and rectifier stage enables practical utilization of the inverter output for DC applications. The results demonstrate that:

- The transformer effectively isolates and scales the voltage
- The rectifier converts high-frequency AC into DC
- The output across the load is a smoothed DC voltage with minimal ripple

This configuration validates the applicability of the Class EF2 inverter in real-world power conversion systems such as DC power supplies and battery charging applications.

E. Output Voltage Waveform (V_{dc})

The output voltage waveform obtained across the $5\,\Omega$ load after rectification is shown in Fig. 6.

It is observed that the output is a steady DC voltage.

XII. Conclusion

In this work, a Class EF2 inverter was successfully designed, analyzed, and simulated. The mathematical modeling provided a clear understanding of the operating principles and enabled accurate determination of component values.

The simulation results verified the theoretical analysis, demonstrating proper inverter operation and desired waveform characteristics. The addition of a transformer and rectifier stage further validated the practical applicability of the system by producing a steady DC output across the load.

Overall, the Class EF2 inverter offers improved performance in terms of efficiency and reduced switching stress, making it a suitable choice for high-frequency power conversion applications.

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